Lab 4: PID Tuning - Open-Loop Transient Response Method

1. Objective

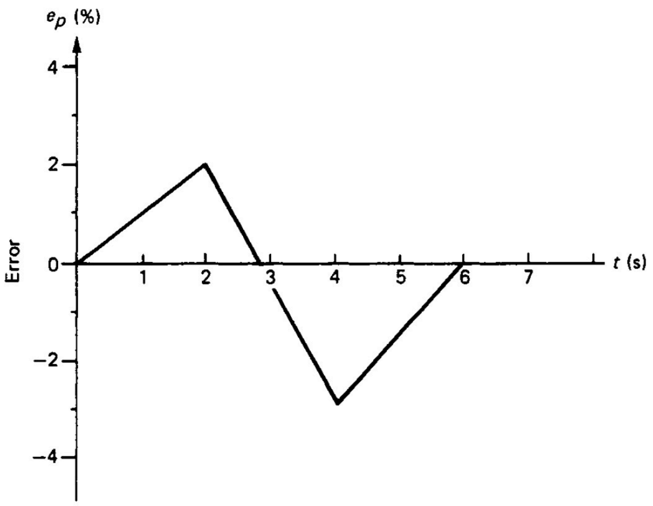
The goal of this lab is to determine the PID parameters (K_p, T_i, T_d) of a temperature control system using the **Open-Loop Transient Response Method** (also known as the **Process Reaction Curve Method**).

2. Pre-lab Task (10 marks)

Instructions: Complete these tasks before your scheduled lab session. Show your working clearly.

2.1 Proportional Controller Analysis (5 Marks)

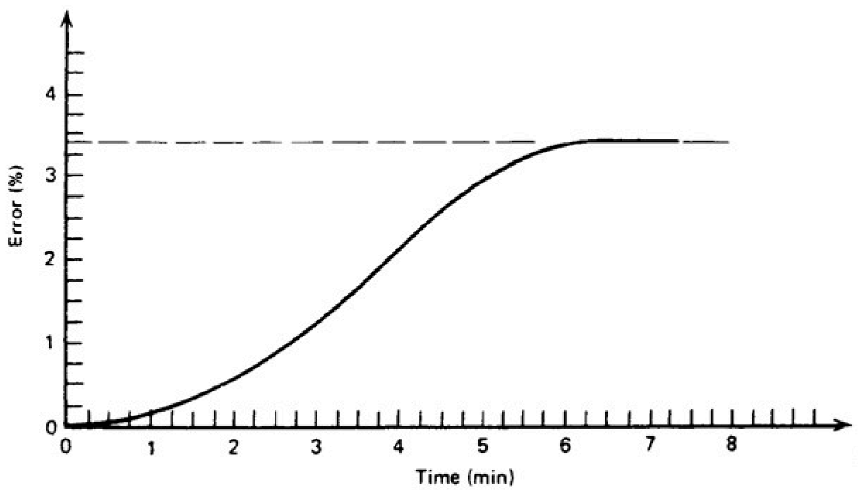
A proportional controller has a proportional gain, $K = 2.0$, and a bias, $p = 0.5$.
Determine the controller output (P) over time when the error signal shown below is inputted to the controller.



Recall the formula: $P = K \cdot e + p$

2.2 Theoretical ZN Tuning (5 Marks)

An open-loop step response of a process is given in the figure below when a step change of **7.5%** in the control variable (MV) is applied.

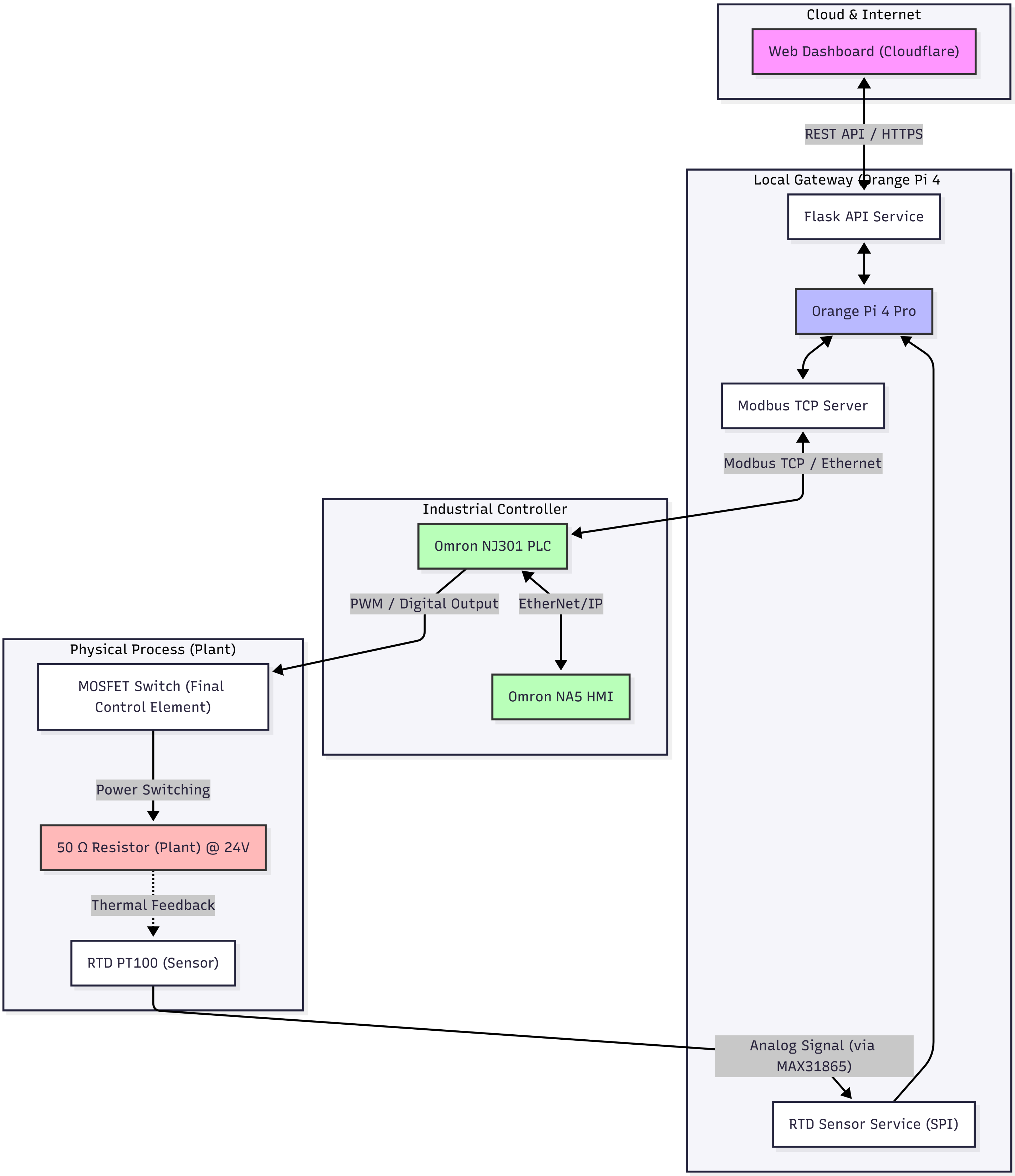


Using the Open-Loop Transient Response Method, determine the appropriate parameters for a **PID Controller** (i.e., K_p, T_i, and T_d).

3. Background

3.1 Lab Setup & Remote Architecture

Before starting the experiment, it is important to understand how your commands reach the physical hardware in the laboratory. This remote setup allows you to interact with industrial-grade equipment from anywhere.



System Components:

- **Web Dashboard**: Your user interface for real-time monitoring and control.
- **Cloud Worker**: A secure bridge that routes your dashboard commands.

- **Orange Pi Gateway**: Centrally manages data flow locally.
- **Hardware Core**: Omron NJ301 PLC, Radxa Camera, and ESP32 Smart Relay.

3.2 The Open-Loop Technique

In an open-loop test, the controller is set to **Manual Mode**. A sudden step change is applied to the Manipulated Variable (MV), and the resulting response of the Process Variable (PV) is recorded.

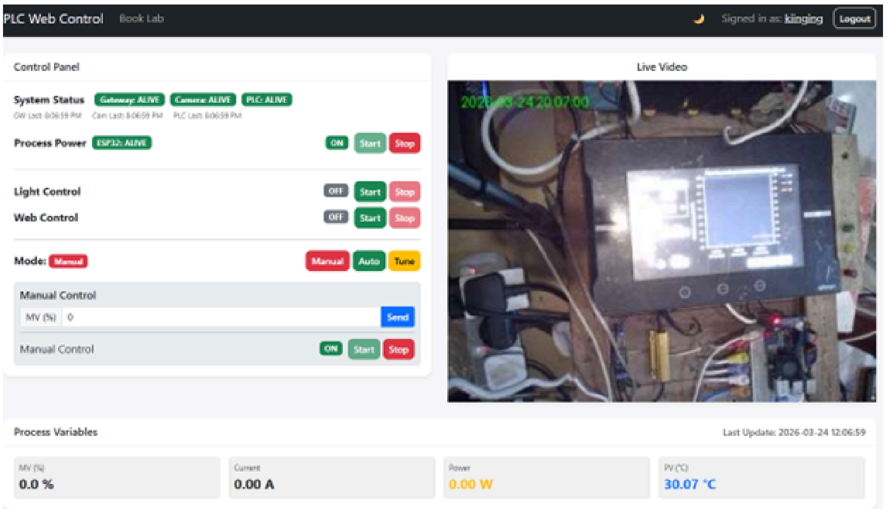
3.3 Understanding Process Gain (K)

- **0% MV** leads to a steady-state temperature of **27°C** (Ambient).
- **100% MV** leads to a steady-state temperature of **130°C**.
- **Process Gain (K)**: $K = \frac{\Delta PV}{\Delta MV}$
- Knowing the full range (**103°C**) helps in normalizing the response.

4. Measurement Tasks

4.1 System Readiness Check

1. **Dashboard Overview**: Familiarize yourself with the interface shown below.



2. **Verify Connectivity**: Look at the **System Status** badges. Both **Gateway** and **ESP32** must be **ALIVE**.
3. **Troubleshooting**: Contact: `wong.kiing.ing@curtin.edu.my` or WhatsApp `0128789001` .

4.2 Powering Up

1. **Process Power**: Click **Start** in the Process Power section.
2. **Camera**: It takes ~1 minute for the camera to come online.
3. **Light Test**: Toggle the **Light Control** while watching the video to confirm real-time control.

4.3 Thermal Experiment

1. **Web Control**: Click **Start** to enable PLC communication.
2. **Manual Initialization**: Set mode to **Manual** and MV to **40%**.
3. **Steady State**: Wait for the temperature to stabilize at 40% MV.
4. **The Step Change**: Change MV to **45%** (a 5% step) and click **Start**.
5. **Data Export**: Wait for stabilization at 45% MV, then click **CSV** to download the data.

4.4 PID Controller Verification (Auto Mode)

1. **Parameter Entry**: Switch the system to **Auto Mode**.
2. **Input Tuning Values**: Enter your calculated PB, T_i, and T_d values into the PID controller fields on the dashboard.
3. **Initial Stabilization**: Set the **Set-point (SP)** to **50°C** and wait for the PV to reach and settle at this value.
4. **Transient Response Test**: Once stable, change the **Set-point (SP)** to **60°C** and click **Start**.

5. **Record Performance**: Observe the temperature graph. Ensure you capture the entire transition from 50°C to 60°C. Download the **CSV** once the system has stabilized at the new set-point.

5. Post Laboratory Task (10 marks)

5.1 Processing the recorded data (5 Marks)

1. **Time Conversion**: Open your CSV in Excel. Column A contains the timestamps. To calculate elapsed time in seconds, subtract the first timestamp from the current one and multiply by 86,400 (the number of seconds in a day).
   - *Example Formula in Column E:* `=(A2 - $A$2) * 86400`
2. **Signal Smoothing**: The temperature sensor is sensitive. Use a 100-sample moving average on the **PV (Column B)** to smooth the signal for your analysis.
   - *Example Formula in Column F:* `=AVERAGE(B2:B101)`
3. **Trend Plotting**: Create a single line chart showing both the **Temperature (PV)** and the **Manipulated Variable (MV)** from Column D.
   - *Pro-Tip*: Since MV is 0-100% and Temperature can reach 130°C, use a **Secondary Y-Axis** for the Temperature to ensure both curves are clearly visible.

5.2 Determination of PID control parameters (5 Marks)

1. **S-Curve Analysis**: Using your smoothed plot, identify the inflection point. Draw a tangent line to determine the **Dead Time (L)** and the **Reaction Rate (N)**.
2. **System Response**: Express the change in temperature (ΔPV) and the change in power (ΔMV) as percentages of their full ranges (103°C and 100% respectively).
3. **ZN Tuning**: Use the Ziegler-Nichols Open-Loop formulas to calculate the final values for K_p, T_i, and T_d.

5.3 Transient Response Analysis (5 Marks)

1. **Plotting the Step Change**: Using the data from Task 4.4, plot the **PV (Temperature)** and **SP (Set-point)** on the same graph as the system moves from 50°C to 60°C.
2. **Performance Metrics**: On your plot, identify and label the following:
   - **Overshoot**: The maximum value minus the set-point.
   - **Rise Time**: Time taken for the temperature to first reach 60°C.
   - **Settling Time**: Time taken for the response to stay within ±2% of the final value.
3. **Discussion**: Compare the observed response with what you expected from your ZN parameters. Is the system stable? Is the overshoot acceptable?

6. Omron NJ301 PLC Implementation

6.1 PLC Conversion

The PLC uses **Proportional Band (PB)**. Convert your calculated K_p using:

$$PB = \frac{100}{K_p}$$

6.2 Summary Checklist

- [ ] Reach steady state at 40% MV.
- [ ] Step to 45% MV and capture S-curve data.
- [ ] Calculate K_p, T_i, T_d and convert to PB, T_i, T_d.
- [ ] In **Auto Mode**, stabilize at 50°C SP.
- [ ] Perform SP step change to 60°C and record transient response.